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On the Semantics of Replacement

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The topic

Expressions and constructions of replacement:

- widespread across languages
- of considerable interest syntactically, semantically, and philosophically.

They seem to so far have escaped the attention of linguists and philosophers.

Types of expressions of replacement in English:

- Verbs of replacement: *replace*, *exchange*
- *Instead*-phrases and the adverb *instead*
- *in place*-phrases (in your place)

The aim of the talk

- Present syntactic and semantic generalizations about of different replacement expressions in English
 - Outline a unified semantics expressions of replacement based on the notion of a structured whole
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1. Replacement verbs

Replace applies to an entity relative to a contextually given *structured whole*:

An artefact:

(1) a. John replaced the leg (of the table).

A spatial region:

- (1) b. John replaced the picture on the wall.

A social institution:

- (1) c. John replaced the chairman (of the company).

A planned event (Ardisson et al. 2025):

- (1) d. John replaced the priest (for the ceremony).

An individual with their alienable possessions:

- (1) e. John replaced his computer / his trainer / his glasses.

Container with content:

- (1) d. Joe replaced the water in the vase.

Temporary configuration:

- (1) f. Mary replaced the bag on the desk.

The constraint on structured wholes:

- (2) a. ??? Sue replaced the flower she had just picked.

b. Sue replaced the orchid in the flower arrangement.

- (3) a.??? John replaced the person speaking to his wife.

b. John replaced the person chairing the session.

- (4) a. ??? John replaced the person that was spontaneously addressing the crowd.

b. John replaced the first speaker at the conference.

Replace targets structured wholes that come with a particular role, place, position or *slot*

A slot: can be filled by different entities

Replacement:

mapping of a structured whole with a slot occupied by a *replacee* and then the *replacer*.

Replacement can also target roles without specific particular wholes

- (5) a. John replaced the housekeeper.

b. John replaced his umbrella.

Connection to an independent entity is required – Langacker’s (1993) ‘reference point’:

Have-relation and possessive construction: the independent entity ‘has’ the replace / replacer.

- (5) a’. John’s umbrella

b.' the umbrella John had

No replacement of locations

Replacement targets a part of a structured entity that occupies particular slot *within* that entity.

Replace cannot target the temporal or spatial location of the entity itself: `

(6) a. ???John replaced the location of the company.

b. ??? Mary replaced the time of the ceremony.

Entities have a location in time and space, but that location itself does not occupy a slot within the entity.

What is a slot?

Potential candidates:

1. Fine's (1999) notion of variable embodiment

An entity associated with a function mapping a time (or circumstance- to manifestations of that entity

Observation: no constraint on that function.

2. Slots in slot mereology (Bennett 2013)

Slot mereology originally developed for abstract wholes such as situations and structural universals, which permit the same entity occurring as a part multiple times

The view:

An entity may be composed of slots, so that the part relation P is relativized to a slot

Observation: no constraints on slots.

What underlies a slot?

Property or relation applicable to an object

Positionalism in the metaphysics of relations (Fine 2000)

(Proper) relations are associated with distinct positions, though without direction or order

In linguistics:

Relational predicates are associated with theta roles (without order among them), to be assigned to occurrences of expressions in the sentence.

Critique of positionalism (Fine 2000):

What are positions ontologically?

Response: they are on a par with roles, and we accept roles in our ontology explicitly

Also: positions are needed:

- Relations may be conjoined with respect to particular positions ('theta-identification')
Slots may consist in the 'identification' of positions of different relations or properties
- Positions are needed linguistically (theta-roles)

Structured wholes

Aristotle: wholes are composed of matter and form

Form: various essential relations or properties that obtain among or of parts

Replacement:

Targets slots whose fillers are not themselves essential to the whole.

In Fine's (1999) terms:

Replacement targets rigid embodiments consisting of variable embodiments.

(7) Replacement applied to structured entities

For a structured whole w with a slot l occupied by an object a , for an event e and agent d ,
and an object b ,
if $\text{REPLACE}(e, d, a, b, l, w)$, then a occupies l at the outset of e and b occupies l as a result
of e .

The verb *exchange*:

Sometimes interchangeable with *replace*:

(8) a. John replaced the screw.

b. John exchanged the screw.

Sometimes not:

(9) a. John exchanged the speakers.

b. John replaced the speakers.

(10) a. John and Mary exchanged addresses.

b. John has Mary's address.

c. Mary has John's address.

Exchange: targets entities a and b , occupying distinct slots

Replace: targets entities for a single slot.

2. *Instead*-phrases

Four types of *instead*-phrases

2.1. *Instead*-phrases of type 1

- take various complements (NPs, APs, PPs, gerunds)
- relate to particular constituents in the sentences by *association with focus*:

- (11) a. Instead of *Mary*, Bill met *Sue*.
 b. Instead of a *man* the firm hired a *woman*.
 c. Instead of *running*, John *walked* to the house.
 d. Instead of *over* the house, the bird flew *into* the house.
 e. Instead of giving *John a book*, Mary gave *Bill a pen*.
 f. Instead of *the children*, Mary met with *the parents*.

Syntax of type 1 *instead*-phrases

They seem to engage in various constructions:

Subordination (PPs):

- (12) a. In place of John, they hired Mary.

Comparatives:

- (12) b. Rather than walking to the house, John ran toward it.

Coordination:

- (12) c. The bird flew into rather than /instead of over the house.

Instead-phrase also apply to structured wholes with slots: structured propositions

Slots in such structured propositions

The positions in the structured proposition filled by the semantic values of the focused constituents.

Replacement by *instead*-phrases

Using alternative semantics (Rooth 1993) for the semantics of *instead*-phrases:

Instead involves mapping the focus-semantic value of the sentence applied to the replacee to the focus-semantic value of the sentence applied to the replacer.

Higher-order semantic values and plural semantic values as replacers

Replacer/ replacee may be

- an object (11a)
- a higher-order semantic value, e.g., a quantifier (11b) or a relation (11c, d)
- an ordered plurality of objects, relating to distinct positions in the structured proposition at once (11e)
- an unordered plurality (11f)

Instead-phrases, unlike *replace*, may relate to spatial and temporal locations:

(13) a. Instead of *Monday*, they arrived *Tuesday*.

b. Instead of in the *garden*, they celebrated in the *house*.

Reason:

Temporal and spatial locations may occupy places in a structured proposition; they are not locations of structured propositions.

2.2. *Instead*-phrases of type 2

Instead-phrases that

- take a gerund or, in, e.g., German, a *that*-clause
- do not associate with focus and relate to particular positions in the sentence
- relate to a slot-providing situation that is entirely given by the context (background situation).

(14) a. Instead of raining, it snowed.

b. Instead people dancing in the street, there was silence everywhere.

c. Instead of John praising himself, Joe criticized Mary.

(15) Statt dass es regnete, schneite es.

instead that it snowed, it rained

Background situation

contains a slot that was filled by an event expected to occur and that is filled by an alternative event.

Type 2 *instead*-sentences are restricted to event descriptions (?):

- (16) a. ???Instead of knowing / speaking French, John knows / speaks Spanish.
 b. Instead of speaking French, John spoke Spanish during the meeting.

Background situations as a structured whole containing a slot:

Slot here individuated by spatio-temporal location and concurrence with other events or states.

What distinguishes structured propositions and background situations from entities that are structured wholes ?

The latter are generated by semantic or pragmatic processes; the former is part of our ordinary ontology.

Structured propositions: result semantic composition (of focus-semantic values)

Background situations: result of pragmatic processes (serving enrichment)

Toward a formal semantics of *instead*

- (17) a. With association with focus

$[Instead\ of\ X_F [\dots Y_F \dots]_S]^w = 1$ iff for the focus-semantic value of p of S , $p^w([Y]) = 1$,
 provided that $p^w([X]) = 0$ and for any world w' in the common ground, $p^{w'}([X]) = 1$.

- b. Without association with focus

For a contextually given situation c with a slot l for a subsituation s ($c(l/s)$),
 $[instead\ of\ S', S] = 1$ iff there is a situation s' , $s' < w$, $s' \in pos(S)$, $\lambda s[c(l/s)](s') = 1$,
 provided $s \notin pos(S)$ and there is a possible situation s'' , $s'' \in pos(S')$,
 and $\lambda s[c(l/s)](s'') = 1$.

$pos(S)$: the situations described by S (the exact truthmakers of S).

2.3. The anaphoric adverb *instead*

Instead without complement when associating with focus:

acts as an adverbial that, relates anaphorically to the replacee in a preceding negative sentence:

- (18) a. Bill did not meet *Mary*. Instead, he met *Sue*.
 b. The firm should have hired a *woman*. Instead, it hired a *man*.
 c. John did not *run* to the use. Instead, he *walked* to the house
 d. The bird did not fly *over* the house. Instead, it flew *into* the house.
 e. Mary did not give *John a book*. Instead, she gave *Bill a pen*.

The adverb *instead* in a sentence *S* requires the preceding sentence *S'* to entail or implicate the negation of what *S* describes applied to the replacee rather than the replacer.

Instead without complement when not associating with focus:

Only requires the preceding sentence to entail or implicate the negation of *S*:

- (19) a. It should have rained. Instead, it snowed.
 b. People did not dance in the street. Instead, there was silence everywhere.
 c. John did not praise himself. Instead, Joe criticized Mary.
 d. ? John does not know French. Instead, he knows Spanish.

Semantics

Instead takes a contextually given proposition as an argument,

Same semantics as *instead*-phrases:

- (20) For a contextually given situation *c* with a slot *l* for a subsituation and a proposition *p* (set of situations) that *instead* refers to,
 $[Instead\ S] = 1$ iff there is a situation s' , $s' < w$, $s' \in pos(S)$, $\lambda s[c(l/s)](s') = 1$, provided
 $S' \notin pos(S)$ and there is a possible situation s'' , $s'' \in p$ and $\lambda s[c(l/s)](s'') = 1$

2.4. Replacement with descriptions

Instead-phrases in descriptions require a different semantics:

- (21) a. the student that uses a typewriter instead of a computer
 b. the house that has a balcony instead of a garden
 c. the spoon that is made of wood instead of silver
 d. the bird that flew into instead of over the house

The abstract wholes on which *instead*-phrases can operate may be expected types, as in (21), kinds as in (22a) as well as previous or future types of stages of objects as in (22b, c):

- (22) a. the swans that are black instead of white
 b. the house that now has two doors instead of one
 c. the house that will be blue rather than yellow

Towards a semantics

Focus-semantic value:

Function from semantic values to descriptions of types of objects or descriptions of objects

Background type (with replacee) mapped onto description (with replacer) of individuals

4. *Place*-NPs

Place-phrases

- come with a counterfactual conditional force
- generally relate to the subject of the sentence:

- (24) a. In your place, I would have declined the offer.
 b. ??In your place, the company would have hired Bill.

Place-NPs involve a slot in a contextually given situation

Connection between antecedent and consequent established on the basis of empathy or simulation.

Equivalence to the antecedent of empathetic conditionals:

- (25) If I were you, I would have declined the offer.

‘You’ refers to the addressee as someone occupying the slot in the contextually given situation.

References

- Ardisson, P, F. Dikme,, F. Moltmann, K. Mueller (2025): ‘The Role of Plans for Qua Events and their Effect on Part Structure’. *Proceedings of Formal Ontology in Information Systems* (FOIS) 15, IOS Press, Amsterdam.
- Bennett, K. (2013): ‘Having a Part Twice Over’. *The Australasian Journal of Philosophy* 91, 83-103
- Langacker, R. W. (1993): ‘Reference Point Constructions’. *Cognitive Linguistics* 4.1., 1-38.
- Fine, K. (1999): ‘Things and Their Parts’. *Midwest Studies in Philosophy* 23(1), 61–74.
- Fine, K. (2000): ‘Neutral Relations’. *Philosophical Review* 109.1., 1-33.
- Rooth, M. (1992): ‘A Theory of Focus Interpretation’. *Natural Language Semantics* 1, 75–116.